

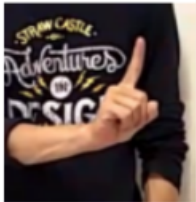
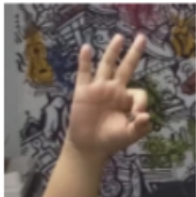
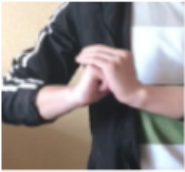
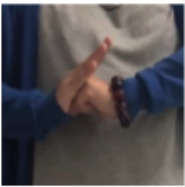
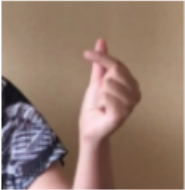
Gesture Recognition

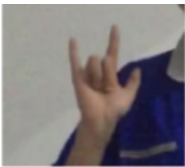
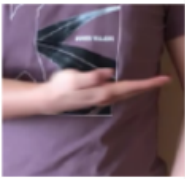
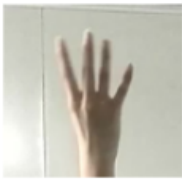
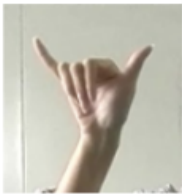
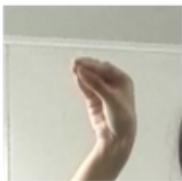
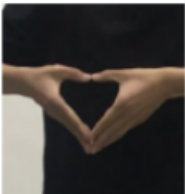
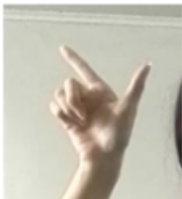
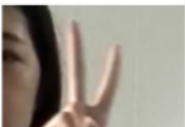
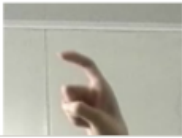
1. Gesture Recognition Overview

1. This gesture recognition routine is based on Baidu Intelligent Cloud platform services, which offers 50,000 free uses per day for learning purposes only. Do not use for commercial purposes. If long-term use is needed, please purchase the relevant service. Our company is not responsible for any issues.

2. API Application Instructions. Official API address:
<https://cloud.baidu.com/doc/BODY/s/ajwvxyyn0>. Log in with your Baidu account after opening the official API address in a browser. If you don't have a Baidu account, please register first. Overseas users need to contact technical support.

1. Supported gestures and example images:

Serial number	Gestures name	Sample
1	number_1	
2	number_5	
3	fist	
4	ok	
5	pray	
6	congratulation	
7	honour	
8	heart_single	
9	thumb_up	
10	thumb_down	

11	i_love_you		17	number_3	
12	palm_up		18	number_4	
13	heart_1		19	number_6	
14	heart_2		20	number_7	
15	heart_3		21	number_8	
16	number_2		22	number_9	

23	rock	
----	------	--

2. API Function. If recognition fails, **apply for your own secret key and replace APP_ID, API_KEY, SECRET_KEY.**

```

from aip import AipBodyAnalysis
""" Your APPID AK SK """
APP_ID = 'your App ID'
API_KEY = 'your Api Key'
SECRET_KEY = 'your Secret Key'
client = AipBodyAnalysis(APP_ID, API_KEY, SECRET_KEY)
""" Read image """
def get_file_content(filePath):
    with open(filePath, 'rb') as fp:
        return fp.read()
image = get_file_content('example.jpg')
""" Call gesture recognition """
Res = client.gesture(image);

```

Gesture Recognition Return Example:

```

{
  "log_id": 4466502370458351471,
  "result_num": 2,
  "result": [{
    "probability": 0.9844077229499817,
    "top": 20,
    "height": 156,
    "classname": "Face",
    "width": 116,
    "left": 173
  },
  {
    "probability": 0.4679304957389832,
    "top": 157,
    "height": 106,
    "classname": "Heart_2",
    "width": 177,
    "left": 183
  }
  ]
}

```

2. Main Code

Code path: /home/yahboom/Dofbot/6.AI_Visual/4.Gesture recognition.ipynb

Import Baidu API, important: change the secret key to your own.

```

import cv2
import time
import demjson
import pygame
from aip import AipBodyAnalysis
from aip import AipSpeech
from PIL import Image, ImageDraw, ImageFont
import numpy
import ipywidgets.widgets as widgets

# For specific gestures, please refer to official information
https://ai.baidu.com/ai-doc/BODY/4k3cpywrv

```

```
#Please refer to the official information for specific gestures
https://ai.baidu.com/ai-doc/BODY/4k3cpywrv
hand={'One':'Number 1','Two':'Number 2','Three':'Number 3','Four':'Number 4',
      'Five':'Number 5', 'Six':'Number 6','Seven':'Number 7',
      'Eight':'Number 8','Nine':'Number 9','Fist':'Fist','Ok':'OK',
      'Prayer':'Prayer','Congratulation':'Congregation','Honour':'Farewell',
      'Heart_single':'Heart','Thumb_up':'Like','Thumb_down':'Dislike',
      'ILY':'I Love You','Palm_up':'Palm Up','Heart_1':'Double Heart 1',
      'Heart_2':'Double Heart 2','Heart_3':'Double Heart 3','Rock':'Rock',
      'Insult':'Middle Finger','Face':'Face'}
```

```
# The key below needs to be replaced with one's own
```

```
#The key below needs to be replaced with one's own
```

```
""" Body Analysis APPID AK SK """
```

```
APP_ID = '31069241'
```

```
API_KEY = 'pxVueLwdAGX4dafYeLsLdZa1'
```

```
SECRET_KEY = 'VsDmfGRlWGqzGhcWowoCT5km4TG4Gylq'
```

```
client = AipBodyAnalysis(APP_ID, API_KEY, SECRET_KEY)
```

```
g_camera = cv2.VideoCapture(0)
```

```
g_camera.set(3, 640)
```

```
g_camera.set(4, 480)
```

```
g_camera.set(5, 30) #Set frame rate
```

```
g_camera.set(cv2.CAP_PROP_FOURCC, cv2.VideoWriter_fourcc('M', 'J', 'P', 'G'))
```

```
g_camera.set(cv2.CAP_PROP_BRIGHTNESS, 40) #Set brightness -64 - 64 0.0
```

```
g_camera.set(cv2.CAP_PROP_CONTRAST, 50) #Set contrast -64 - 64 2.0
```

```
g_camera.set(cv2.CAP_PROP_EXPOSURE, 156) #Set exposure 1.0 - 5000 156.0
```

```
ret, frame = g_camera.read()
```

Camera display component

```
image_widget = widgets.Image(format='jpeg', width=600, height=500) #Set camera
display component
display(image_widget)
image_widget.value = bgr8_to_jpeg(frame)
```

Main display and result program.

```
try:

    while True:
        """1. Capture """
        ret, frame = g_camera.read()

        #image = get_file_content('./image.jpg')

        """ 2. Call gesture recognition """
        raw = str(client.gesture(image_widget.value))
        text = demjson.decode(raw)
        try:
            res = text['result'][0]['classname']
```

```

except:
    #         print('Recognition result: Nothing detected~' )
    #         img = cv2ImgAddText(frame, "Not recognized", 250, 30, (0, 0 , 255),
30)
        img = frame
    else:
        #         print('Recognition result: ' + hand[res])
        #         img = cv2ImgAddText(frame, hand[res], 250, 30, (0, 255 , 0), 30)
        if res == 'Prayer': # 1 Prayer
            print('Recognition result: ' + hand[res])
            img = cv2ImgAddText(frame, hand[res], 250, 30, (0, 255 , 0), 30)

        elif res == 'Thumb_up':# 2 Like
            print('Recognition result: ' + hand[res])
            img = cv2ImgAddText(frame, hand[res], 250, 30, (0, 255 , 0), 30)

        elif res == 'Ok': # 3 OK
            print('Recognition result: ' + hand[res])
            img = cv2ImgAddText(frame, hand[res], 250, 30, (0, 255 , 0), 30)

        elif res == 'Heart_single': # 4 Single hand heart
            print('Recognition result: ' + hand[res])
            img = cv2ImgAddText(frame, hand[res], 250, 30, (0, 255 , 0), 30)

        elif res == 'Five': # 5 Number 5
            print('Recognition result: ' + hand[res])
            img = cv2ImgAddText(frame, hand[res], 250, 30, (0, 255 , 0), 30)

        elif res == "Eight": # Number 8
            print('Recognition result: ' + hand[res])
            img = cv2ImgAddText(frame, hand[res], 250, 30, (0, 255 , 0), 30)

        elif res == "Rock": # rock
            print('Recognition result: ' + hand[res])
            img = cv2ImgAddText(frame, hand[res], 250, 30, (0, 255 , 0), 30)

        elif res == "Congratulation": # Congregation
            print('Recognition result: ' + hand[res])
            img = cv2ImgAddText(frame, hand[res], 250, 30, (0, 255 , 0), 30)

        elif res == "Seven": # Number 7
            print('Recognition result: ' + hand[res])
            img = cv2ImgAddText(frame, hand[res], 250, 30, (0, 255 , 0), 30)

        elif res == "Thumb_down": # Thumb down
            print('Recognition result: ' + hand[res])
            img = cv2ImgAddText(frame, hand[res], 250, 30, (0, 255 , 0), 30)

        else:
            img = frame
            image_widget.value = bgr8_to_jpeg(img)
except KeyboardInterrupt:
    print(" Program closed! ")
    pass

```

After running the code blocks, you can see the recognized gestures.

